Notes on GCN Theory

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1. Literature Review

1.1. Expressiveness

We first review papers on the expressiveness of GCNs. Note that the expressiveness is measured by the training loss, and the theory is similar to the classic theory showing that a feed-forward neural network can approximate any continuous function within any accuracy. However, due to the weight-sharing mechanism, the analysis is different.

Before proceeding, we introduce two terms: *Separation power* means a network's ability to distinguish two different inputs, e.g., generating different outputs; *Representation power* means its ability to approximate functions of interest. In the following works, the authors first prove the separation power, and then the representation power is easy to derive.

We will consider the following mixed-integer linear program:

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} \circ \boldsymbol{b} \\ & \boldsymbol{l} \le \boldsymbol{x} \le \boldsymbol{u} \\ & x_j \in \mathbb{Z} & \forall j \in S, \end{aligned}$$

where $\circ_i \in \{\geq, =, \leq\}$. The MILPs are represented by m constraint nodes with features $h_i^C = (\circ_i, b_i)$, n variable nodes with features $h_j^V = (c_j, l_j, u_j, \mathbb{1}_{j\in S})$, and mn connecting edges with weights (A_{ij}).

Chen et al. (2023a) investigate whether a specific form of GCN can predict the feasibility, boundedness and an optimal solution to a linear program. They mainly utilize the Weisfeiler-Leman (WL) graph isomorphism test, which is proved to have the same separation power as GCNs (see Xu et al. 2019). Under the WL test, the constraint node set $[m]$ and the variable node set $[n]$ will be partitioned into some disjoint sets $[m] = I_1 \cup I_2 \cup \dots I_s$ and $[n] = J_1 \cup J_2 \cup \dots J_t$ such that

1. $h_i^C = h_{i'}^C$ if $i, i' \in I_p$ for some $p \in [s]$;
2. $h_j^V = h_{j'}^V$ if $j, j' \in J_q$ for some $q \in [t]$;
3. For any $p \in [s]$, $q \in [t]$ and $i, i' \in I_p$, we have $\sum_{j\in J_q} A_{ij} = \sum_{j\in J_q} A_{i'j}$;
4. For any $p \in [s]$, $q \in [t]$ and $j, j' \in J_q$, we have $\sum_{i\in I_p} A_{ij} = \sum_{i\in I_p} A_{ij'}$.

In this case, if the WL test cannot distinguish two LP instances G and $\hat{G}$, then we can deduce that their resulting separation sets are the same and the parameters satisfy the following conditions

1. $h_i^C = \hat{h}_i^C$ and $h_j^V = \hat{h}_j^V$;
2. For any $q \in [t]$ and any $i \in [m]$, we have $\sum_{j\in J_q} A_{ij} = \sum_{j\in J_q} \hat{A}_{ij}$;
3. For any $p \in [s]$ and any $j \in [n]$, we have $\sum_{i\in I_p} A_{ij} = \sum_{i\in I_p} \hat{A}_{ij}$.

In this case, G and $\hat{G}$ share the same feasibility and optimal objective value because given a feasible solution of either one, we can construct a feasible solution to the other one. Note that the constructed solution is of the form $\hat{x}_j = \frac{1}{|J_q|}\sum_{j\in J_q} x_j$ for any $j \in J_q$, and hence the proof cannot hold when there are integral variables.

As for the optimal solution, given that optimal solutions may not be unique, we first restrict our attention to the optimal solution with the smallest L_2-norm, which is unique. (Note that the uniqueness does not hold when there are integral variables.) Then, we can prove that the two LP instances share the same optimal solution.

Chen et al. (2023b) investigate the same problem for MILPs. In this case, due to the integral variables, the problem is significantly complicated. First, in this case, there exist two instances having different feasibility statues but the same WL results (see Lemma 3.2). Second, the optimal solution may not be unique if we consider the problem $\max_{x_1,x_2\in\{0,1\}} x_1 + x_2 \quad \text{s.t. } x_1 + x_2 = 1$. In this case, they provide a sufficient "unfoldable" condition: The resulting separation sets satisfy that $|I_p| = 1$ for any $p \in [s]$ and $|J_q| = 1$ for any $q \in [t]$. In other words, the characterization of each node is unique. Under the unfoldable condition, the optimal solution is unique, and the GCN can express all three features of MILPs. Moreover, they also propose some ideas to improve the expressiveness for the foldable instances. Specifically, they show that if we can add some heterogeneous coefficients in each feature vector, GCNs can represent these foldable instances. However, the proposed approaches either cannot scale up or encounter difficulties in the training process.

Chen et al. (2024) investigate whether GCNs can express the strong branching (SB) indices for MILPs. In this case, the SB indices are inherently unique, and the authors define the "message-passing-tractable (MP-tractable)" condition, which is weaker than the unfoldable assumption. Specifically, under the MP-tractable condition, for any $p \in [s]$ and $q \in [t]$, we have $A_{ij} = A_{i'j'}$ if $i, i' \in I_p$ and $j, j' \in J_q$. Given a finite sample set satisfying the MP-tractable condition, the GCN can represent the SB indices to any accuracy. Moreover, they show that a modified GCN that computes pair-wise information in each round can represent the SB indices without the MP-tractable condition.

Chen et al. (2025) investigate the expressiveness problem for the mixed-integer linearly constrained quadratic program (MI-LCQP). In this case, they show that GCNs can express the feasibility and the optimal objective value under the MP-tractable condition, and can express the optimal solution under the unfoldable condition.

Following the above proofs, we may prove that GCNs can express the feasibility and the optimal objective value under the MP-tractable condition, and can express the optimal solution under the unfoldable condition.

1.2. Size Generalization

In this part, we review papers on GCNs' size generalization ability. Due to the weight-sharing mechanism, we can train a GCN on small samples and then apply it to large samples without retraining the network. In this area, researchers typically assume that the model weights are fixed, and establish some theoretical bounds for the gap between outputs of small and large samples (after expanding them to the same size). If the target solution mapping is transferable and the neural network can learn it well, then these theoretical bounds tell us that the "from small to large" idea performs well.

Ruiz et al. (2020) study a sequence of *deterministic* graphs converging to a graph with uncountable number of nodes and assume that the learnable weights are fixed. Given a graph with n nodes, the authors construct a graph with infinite nodes by splitting the interval $[0,1]$ into n intervals and assigning i-th node's feature vector to all nodes within the i-th interval. Let Y_n (Y, resp.) denote the output of the constructed (original, resp.) graph with infinite nodes. Since we approximate a continuous function with a piecewise constant function, the size generalization error can be proved to be $\|Y - Y_n\| = O(\sqrt{n})$. Moreover, we can use the triangle inequality to derive the bound $\|Y_m - Y_n\| = O(\sqrt{m} + \sqrt{n})$.

Levin et al. (2025) study the case where small samples are generated based on large samples which follow some distribution (see Assumption 4.1). First, they introduce two size embeddings, zero-padding embedding and duplication embedding, such that inputs with different sizes can be compared. Then, they analyze the transferability of GCNs. A GCN is *transferable* if its corresponding functions for different sizes are compatible with each other and locally Lipschitz. Note that the transferability only depends on the network structure and hence is independent of the weights. Given N samples with n nodes, if the GCN is transferable and the learned mapping is Lipschitz, we can prove that the generalization gap is upper bounded by $O(1/\sqrt{N} + R(n))$, which tends to zero as N and n tend to infinity. Similar to Ruiz et al. (2020), the generalization gap is also defined as the difference between the outputs of the same GCN given inputs of different sizes. In Section 5, the authors investigate several existing GCNs and find that some models are not transferable. For example, if we use the duplication embedding and the summation aggregation function (i.e., $\mathrm{Agg}_s(\boldsymbol{x}) = \sum_i x_i$), the GCN is not transferable because increasing the input size will change the scale of the aggregated information. In contrast, the GCN is transferable if we use the average aggregation function (i.e., $\mathrm{Agg}_a(\boldsymbol{x}) = \frac{1}{n}\sum_i x_i$).

For the assortment problem, following the proofs in Levin et al. (2025), we may use the zero-padding embedding and check whether the current model is transferable.

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